

Akash Kesav Shanmugaraj

+65 86246231 | akashkesav3@gmail.com | akash-kesav | AkashKesav | Portfolio

SUMMARY

Fourth-year Smart Manufacturing undergraduate at IIITDM Jabalpur focused on applied machine learning for materials science, with experience in distributed ML systems. Currently researching physics-informed inverse design, trust-minimized training, and semiconductor generative modelling; seeking laboratory internships to strengthen rigorous research experience and preparing for PhD study at leading universities worldwide.

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

Bachelor of Technology in Smart Manufacturing | CGPA: 7.9 / 10

#1 IIIT; Govt.-funded Institute of National Importance

Expected June 2027

Jabalpur, Madhya Pradesh

EXPERIENCE

Rakumura IT Solutions

AI/ML Backend Development Intern (Remote)

Jun 2025 – Nov 2025

Chennai, Tamil Nadu

- Architected **multi-agent AI systems** using **2 orchestration frameworks** (LangChain, LangGraph) and AI-assisted development tools (Claude Code, Codex, GitHub Copilot) for rapid prototyping, automating triage, assignment, and status tracking while **reducing manual coordination overhead** for internal operations teams
- Integrated a **4-system stack** (PostgreSQL, MongoDB, Redis, NATS) for asynchronous stateful execution across internal workflow endpoints; containerized services with Docker for reproducible deployments

IIITDM Jabalpur – Materials Science Lab

Research Intern – Machine Learning (On-site)

Nov 2023 – Apr 2024

Jabalpur, Madhya Pradesh

- Benchmarked **5 ML architectures** (GNNs, XGBoost, Random Forest, DNNs) on semiconductor band-gap prediction, achieving **83% accuracy**
- Designed an automated data-collection and preprocessing pipeline ingesting **1,600+ material records** from the Materials Project API across all 118 elements
- Contributed to research abstract accepted at **Materials Today Conference** (Elsevier, 2025)

CURRENT RESEARCH

Physics-Informed Inverse Design of Mechanical Metamaterials

Co-first-author research with Tejas P. Biradar; supervised by Dr. K. Ponappa

Aug. 2026 – Present

IIITDM Jabalpur

- Investigating inverse design under structural-integrity constraints using a Phys-JEPA-inspired latent formulation and PINN baselines.
- Developing the methodology and validation plan; methods, data, and preliminary results remain confidential pending supervisor approval.

A Trust-Minimized Architecture for Hybrid Distributed Training on Untrusted Networks

Advanced manuscript draft; equal-contributing author

Feb. 2026 – Present

IIITDM Jabalpur

- Authors: Ayush Saxena*; Akash Kesav Shanmugaraj*; Aabhas Malpani; Ashish Singh Parihar (*equal contribution).
- Co-designed a hybrid data/model-parallel framework with one-shot Krum-score filtering, softmax-weighted aggregation, VRF-and-trust validator selection, commit-reveal, progressive penalties, dPKI-authenticated ECDHE, and smart-contract coordination.
- Built ResNet-50/CIFAR-10 experiments: softmax-aggregated data-parallel test accuracy of 88.28%, 85.83%, and 83.09% for $N = 2, 4, 8$; clean synchronous-pipeline test accuracy of 88.25% for $K = 2$ and $K = 4$.
- In the evaluated testbench, the Krum-score filter achieved TPR 1.0/FPR 0.0 on Gaussian and Byzantine-majority attacks ($f = 2/8$), where K-means collapsed; VRF+Trust reduced validator-selection Gini from 0.875 to 0.113 ± 0.024 .

Semiconductor-Focused MatterGen Fine-Tuning and Evaluation

Confidential continuation of semiconductor inverse-design research

2026 – Present

IIITDM Jabalpur

- Authors: Akash Kesav Shanmugaraj; Ponvijayakanthan K.
- Contributing to MatterGen fine-tuning for semiconductor candidates and building a systematic evaluation harness for reproducible checkpoint comparison and result validation.
- Technical details and results are withheld while experiments are being finalized.

PUBLICATIONS & PATENTS

SECUF-2026 (Invited Talk) – Co-author, “Inverse design of magneto-excitonic semiconductors,” 5th Conf. Physics under Synergetic Extreme Conditions (China).

Materials Today 2025 (Poster) – ML-based material property prediction, Sitges, Spain.

Patent Filed – Co-filed **1 patent application**, under review.

TECHNICAL SKILLS

Languages: Python, Go, SQL

Frameworks: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph

Core ML & AI: Deep Learning, GNNs, NLP, Transformers, RAG, Agentic AI

MLOps & Dev Tools: Docker, REST APIs, Go Fiber, NATS, Git, Linux, CI/CD, Claude Code, Codex, GitHub Copilot

Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, MongoDB, Redis

LEADERSHIP

Team Lead, IIITDM Jabalpur – Led teams of 4–6 peers across **3 projects** in Product Realization, Engineering Design, and Cyber-Physical Production Systems; delivered prototypes through design reviews (grades **A, B+, A+**).